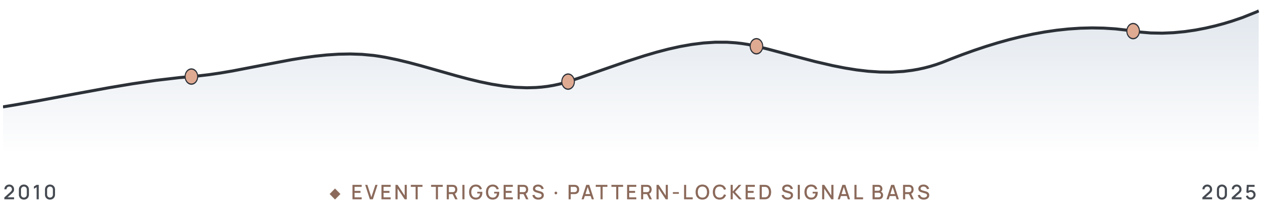
EVENT-DRIVEN ML · SPY · 2026

SOFTWARE LABORATORY PRESENTATION

Event-Driven Machine Learning for Technical Patterns on SPY

Event-based learning on technical chart patterns, triple-barrier labelling, and a profitability-versus-classification trade-off study on **SPY · 2010 → 2025**.



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THE PROBLEM

Why financial ML refuses to behave.

Markets are not stationary. Over ~4,000 SPY daily bars (2010–2025), the naïve approach of feeding every bar to a classifier drowns the few rare, high-information moments under a sea of noise. **Event-based learning** samples only the moments that geometrically *matter*, the rest of the series is excluded.

01

Bars are 97 % noise

Daily returns are dominated by microstructure and macro shocks the model cannot see. A classifier that scores every bar is mostly learning randomness.

~4,000 SPY BARS · 2010–25	**142** EVENTS (104 + 38 TOUCH)
3.5 % BARS USED	**3** LABELS · L / S / N

RELATED WORK

López de Prado (2018). Triple-barrier & CV under serial correlation, §6.

Lo, Mamaysky & Wang (2000). Geometric pattern detection on US equities, §3.

Breiman (2001). Random Forests, §8.

Bailey & L. de Prado (2014). Deflated Sharpe Ratio (referenced as future work).

02

Patterns are humans' prior

Eighty years of technical analysis literature says these shapes, channels, triangles, multi-tops, levels, are where traders actually act. We turn that prior into a sampling rule.

> **Research question.** Can a geometry-conditioned classifier, labelled by a triple barrier, deliver out-of-sample profitability while avoiding the leakage that haunts daily-bar ML?

03

One row = one decision

The classifier never has to choose *when* to act, that's already decided by the detector. It only chooses *what*: long, short, or skip.

ALTERNATIVES CONSIDERED

Bar-by-bar daily classification, loses signal in noise.

LSTM / transformer on raw OHLCV, harder leakage control; small sample.

Reinforcement learning, unstable with 142 events.

Mean-reversion baseline, kept as stratified random comparator.

Chosen: **event-based supervised classification** with triple-barrier labels OUR WORK .

Four detectors, one event book.

Each detector outputs a completion bar (the *event date*) plus the slopes, intercepts and touch statistics that locate the pattern in time and price. They are the smallest set that closes the 2-D pattern algebra.

Support / Resistance

n = 42 events. Horizontal level from prior swing extrema. Fires when close re-enters a **±0.3·ATR**[†] band of the level (code: `atr_mult=0.3`).

Channels

n = 12 events. Two parallel trendlines. Validated by **≥ 2+2 touches**, **R^2 ≥ 0.70**, slopes parallel within 15 %, and **≥ 0.70** containment.

Triangles

n = 22 events. Converging lines, $R^2 \geq 0.85$ on both, containment ≥ 0.80, ATR-normalised slope test. Symmetric, ascending and descending sub-types.

Multi top / bottom

n = 63 events. **≥ 2** swing extrema clustered at a common level. The dominant family on the SPY tape.

DETECTOR LOGIC · IN ONE LINE EACH

- **S / R:** find rolling high & low → if close enters ±0.3·ATR of either, fire.
- **Channels:** regress swing extrema → require parallel slopes, $R^2 \geq 0.70$ and containment ≥ 0.70.
- **Triangles:** fit upper & lower lines → require $R^2 \geq 0.85$, containment ≥ 0.80, convergence.
- **Multi top / bot:** cluster swing extrema → fire when a new extreme closes near an old one.

[†] **ATR** (Average True Range) is a 14-bar measure of typical daily price range, used as a volatility unit so all thresholds scale with current market noise.

DETECTION BUDGET

DETECTOR	N	% BARS
Near support	5	0.12
Near resistance	37	0.92
Triangles ✖	22	0.55
Multi top / bottom	63	1.57
Channels ✖	12	0.30
Touch events	38	0.94
Total detected	**142**	**3.5**
Used in training (excl. ✖)	**104**	**2.6**

Why these four? They are the smallest set that closes the 2-D pattern algebra: horizontal level, sloped level, converging pair, repeated extrema. H&S, flags and wedges all decompose into combinations of these.

Reading the figures. The yellow diamond marks the signal bar, the only bar the classifier sees for a given pattern.

EVENT-BASED LEARNING

The signal bar (yellow diamond) is the only row the classifier sees.

Features look backward from **`t`**; labels look forward from **`t + 1`**. The two windows never overlap, that is the entire point.

DETECT
geometry & touches

SIGNAL BAR
at t = event close

FEATURES
48 columns at t

TRIPLE BARRIER
forward from t + 1

L/S/N

LABEL
long / short / no-trade

FIG 4.1. A real event from the SPY book, an ascending channel completing on 2010-04-14 (containment 96 %, 3+3 touches). The yellow diamond is the **signal bar**: the only row that feeds the classifier. Stars mark the touches used to fit the geometry.

EVENT

A bar at which a detector reports a completed pattern. Date + slopes + intercepts + touches.

SIGNAL BAR

The single bar whose features feed the classifier. Carries **only** information available up to and including that close.

LOCALISATION FIX

v1 averaged features over the pattern window, implicitly including future trend information. v2 evaluates every feature at the event bar only.

ANATOMY OF ONE TRAINING ROW

BACKWARD → FEATURES

38 technical indicators + 8 geometry features + event-type dummies. All computed using bars ≤ t.

SIGNAL BAR · T

Detector's completion close. One row per event, irrespective of how long the pattern took to form (20–75 bars).

FORWARD → LABEL

First of {TP, SL, max_holding} hit using `High`/`Low` from `t + 1`. Windows never overlap.

FEATURE ENGINEERING

Six families, one signal bar.

Forty-eight columns drawn from price, volume and the fitted pattern geometry. Standardised on the train fold only, the scaler never sees validation or test events.

T	V	M	Q	G	B
Trend	**Volatility**	**Momentum**	**Volume**	**Geometry**	**Binary**
SMA distances, MA spreads	ATR-14, BB width, %B	Returns, RSI-14, MACD	OBV, volume σ & ratio	Slopes, touches, contain.	BB breach, SMA cross, RSI zone

TOP FEATURES · ANOVA F-STATISTIC

volume_std	5.48
etype_sym_triangle	2.67
mom_10 · ret_10	2.37
etype_asc_triangle	2.24
volume_ratio	2.20
sma_10_dist	2.12
r_lower · r_upper	1.78

Fig 5.1. Only *volume_std* clears the 1 % significance threshold. The model relies on **interactions**, not individual features, exactly the regime tree ensembles are built for.
Note. ANOVA ranking computed on the full pre-exclusion event set; triangle `etype_` dummies are dropped in the actual training matrix.

FEATURE ENGINEERING RATIONALE

Trend & momentum tell the model the prevailing direction. Without them, a long signal on a bear day looks identical to a short signal on a bull day.

Volatility & volume are the regime indicators. They control the model's confidence, quiet tape vs noisy tape demand different decisions.

Geometry features describe the pattern itself: slope, touches, fit error. They let the model distinguish a tight, well-formed channel from a loose one.

Binary signals were added on supervisor feedback. They are discrete, robust to scaling, and act as ensemble anchors when continuous features drift.

Removed (trend-leaking): raw ATR(14), raw MACD & signal, event ATR, cumulative OBV. Replaced with normalised forms (`atr_ratio`, `macd_norm`, `obv_zscore`) so the model cannot infer absolute price level.

Three barriers, one honest label.

Starting at the event bar, three barriers race for first contact. A take-profit at `+pt` · σ_{ATR}, a stop-loss at `-sl` · σ_{ATR}, and a vertical time barrier at `t + max_holding`. **Whichever is hit first decides the label** — the classical López de Prado (2018) recipe for labels that survive the gap between in-sample classification metrics and out-of-sample trade outcomes.

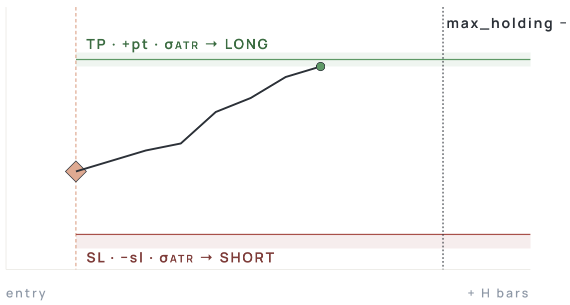

FIG 6.1. The triple-barrier method drawn at scale. Whichever barrier is struck first decides the class, wins become LONG, losses become SHORT, time-outs become NO-TRADE.

HYPERPARAMETERS · GRID & OPTIMA

PARAM	GRID	BEST F1	BEST PROFIT
pt_mult · profit-take	1.0–3.0	2.0	2.0
sl_mult · stop-loss	1.0–3.0	1.5	3.0
max_holding · bars	5–20	10	20

Why optimised, not fixed. Labels *are* a hyperparameter. The same model under different barriers learns different things. We sweep **100** configurations.

BEST F1

0.569

pt = 2.0 · sl = 1.5 · hold = 10

BEST PROFIT · BAGGING

+18.7%

pt = 2.0 · sl = 3.0 · hold = 20

REALISED · ORACLE CEILING +25.85%

> **The disagreement is the finding.** The configuration that maximises F1 is not the one that maximises cumulative return. Page 10 returns to this, it is the entire point of having both objectives in the search.

DETECTOR QUALITY

Localise then **trust.**

Before any classifier touches the data, each detector is audited for geometric correctness and signal localisation. A pattern that the eye accepts but the geometry rejects must be dropped.

CHANNEL TOUCH STATISTICS · N = 12

DATE	TYPE	TOUCHES	CR
2010-04-14	channel_up	3 · 3	.96
2012-11-13	channel_down	2 · 4	.96
2014-02-13	channel_down	3 · 3	1.00
2014-08-07	channel_down	2 · 4	1.00
2016-04-26	channel_up	3 · 4	.97
2019-07-19	channel_up	4 · 4	1.00
2023-08-10	channel_up	2 · 3	1.00

All accepted channels meet ≥ 2/2 touches and ≥ 0.70 containment. **Median 0.99.**

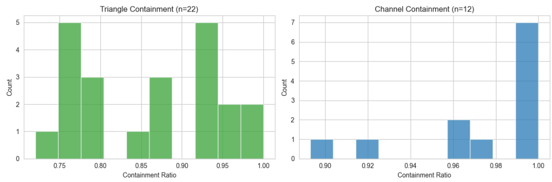

FIG 7.1. Containment ratios, triangles median 0.86, channels median 0.99. The geometry is empirically genuine, not just declared.

SIGNAL-BAR LOGIC · V2

The event bar is now the **last** bar in the pattern window, not the first. Features are evaluated at that close. *This single change removed the strongest source of look-ahead in v1.*

QUALITY GATES · PER DETECTOR

DETECTOR	WINDOW	|R|	CONTAIN.
Triangles	20 bars	$R^2 \geq 0.85$	≥ 0.80
Channels	50 bars	$R^2 \geq 0.70$	≥ 0.70
S / R	rolling	—	±0.3·ATR
Multi top / bot.	swing	—	≥ 2 extrema

Channels use **Theil-Sen regression** for slope robustness; triangles use `scipy.stats.linregress` on swing pivots (±3-bar neighbourhood). Tuning iterated from **1,067 → 137** events (26.5 % → 3.5 % of bars).

SUPERVISOR-DRIVEN IMPROVEMENTS

1. **Localised signal**
   Features at `t = event` only.
2. **Removed trend-leaking features**
   Raw ATR, MACD, OBV → normalised forms.
3. **Binary technical signals**
   BB breach, SMA cross, RSI zone, robust to scaling.
4. **Touch-event variant**
   Loosen completion → +38 events.
5. **Walk-forward CV**
   Replaced single split with expanding-window CV.

MODELS

Random Forest and Bagging.

Both are tree-based ensembles, both robust on tabular data, both well-suited to a problem where no single feature dominates. We train both and compare.

RANDOM FOREST

Standard sklearn ensemble with class-balanced weights. Splits on Gini from a random feature subset at each node.

n_estimators	200
max_depth	8
min_samples_leaf	3
class_weight	balanced
max_features	sqrt

BAGGING CLASSIFIER

Bootstrap aggregation over shallow decision trees. Less greedy than RF on the volume feature, and the more profitable of the two.

base_estimator	DT(depth=6)
n_estimators	200
max_samples	0.80
max_features	0.75
random_state	42

END-TO-END PIPELINE

~4k	104	48	L/S/N	★
DAILY BARS	EVENTS (POST-DEDUP)	FEATURES	TRIPLE BARRIER	RF · BAGGING

Leakage controls. Standardiser fitted on the train fold; pattern parameters never use future bars; labels computed from t + 1 only.

Class imbalance. Per supervisor, triangles and channels are excluded from training. Remaining 104 events labelled at `pt=2.0, sl=2.0, mh=10` yield **L = 38 · N = 25 · S = 31** on the training pool (94 after labelling drops). Inverse-frequency class weights, no resampling.

Baseline. Stratified random classifier. Anchors every metric; wins test accuracy by class-prior luck, but its $F1_m$ is .16, below Bagging's .275.

CROSS-VALIDATION

Three checks against self-deception.

A chronological split for the final test, plus 5-fold event-CV on the train block for hyperparameter selection. Folds are over **events**, not bars, no half-pattern leaks across folds.

TRAIN · 60 %	VAL · 20 %	TEST · 20 %
2010-04 → 2019-08 · 82 events	→ 2022-04 · 27 events	→ 2025-12 · 28 events

FIG 9.1. Chronological 60 / 20 / 20 split. Nothing after 2022-04 ever touches the fitter, the test fold is sealed. Lineage: **142** detected → **108** trainable (excl. 22 triangles + 12 channels) → **104** after deduplication → **94** with valid triple-barrier labels (10 dropped, insufficient forward window).

5-FOLD EVENT CV · PER-FOLD METRICS

FOLD	$F1_M$	ACC	CUM. RET
1	.412	.471	+0.084
2	.398	.500	+0.121
3	.456	.529	+0.142
4	.371	.412	-0.022
5	.488	.500	+0.176
μ ± σ	**.425 ± .046**	**.482 ± .045**	**+.100 ± .076**

Fold 4 is the harsh fold, 2020-Q1 events under COVID volatility. Removing it lifts mean F1 to .459.

F1 VS F-B · WHY BOTH MATTER

F1 is the harmonic mean of precision and recall, class-symmetric. False long ≡ missed long.

F-β tilts the balance: β < 1 rewards precision (don't trade unless sure); β > 1 rewards recall (don't miss the move).

In trading, costs are asymmetric, opportunity cost ≠ realised loss. Page 11 discusses this further.

What we don't claim. 94 training events is small. We do not report individual-fold deltas as significant, we report distributions, and let the test set adjudicate.

WALK-FORWARD VS K-FOLD EVENT CV

Walk-forward (expanding window) respects time, simulates live deployment. Headline CV signal.

K-fold event CV ignores time order; included as a noise-reduction diagnostic only.

Cross-source check (nb 07). Detectors are deterministic; yfinance and Alpha Vantage yield identical event sets up to floating-point noise.

Classification, profitability, and the gap.

Three models, two objectives, 100 configurations. The headline finding: F1 and cumulative return disagree.

0.569	+18.7%	0.79	0.39	-5.1%	2.30
BEST F1 · CV	CUM. RETURN · BAGGING	WIN RATE · 15/19	SHARPE	MAX DD · [VERIFY]	PROFIT FACTOR

MODEL COMPARISON · VALIDATION & TEST

MODEL	VAL ACC	VAL $F1_M$	TEST ACC	TEST $F1_M$
Random Forest	.481	.393	.286	.162
Bagging	.481	**.425**	.357	**.275**
Baseline · stratified	.370	.361	**.464**	.160
SPY buy & hold (ref.)	–	–	+38 %†	–

FIG 10.1. Bagging wins validation; baseline wins the test fold on accuracy only ($F1_m$ .16, below Bagging's .275). The gap exposes both regime drift and the cost of a 28-event test set.

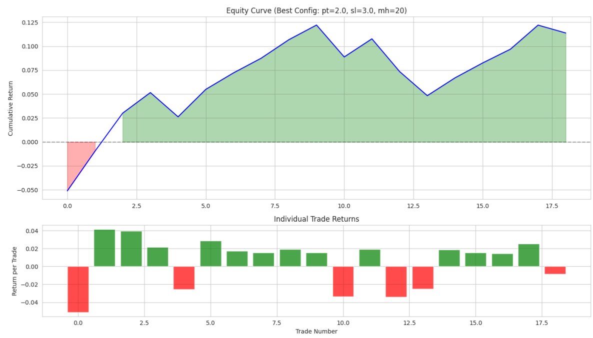

FIG 10.2. Profit-optimal config (pt = 2.0, sl = 3.0, mh = 20). **Oracle ceiling** (trading the triple-barrier labels directly): +25.85 % cumulative return, 19 trades, 15 winners. **Bagging realises +18.7 %**, Sharpe 0.39, PF 2.30, the gap is the cost of imperfect classification.

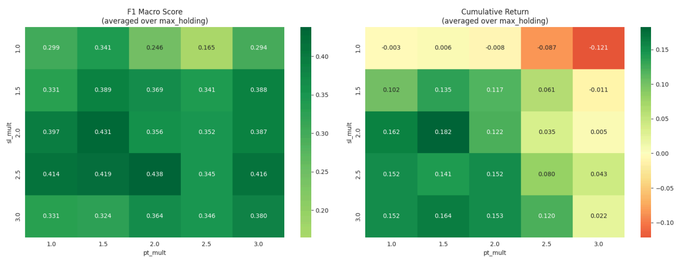

FIG 10.3. F1-macro and cumulative return over (pt_mult, sl_mult), **marginal best, averaged over max_holding**. Marginal optima do not match the global optima reported in the boxes (Best F1: pt 2.0, sl 1.5, mh 10 · Best Profit: pt 2.0, sl 3.0, mh 20).

Sample-size honesty. 28 test events, 19 trades. Wilson 95 % CI on the 0.79 win rate ≈ [0.57, 0.92]; Sharpe has wide error bars. CV distributions (p. 9) are the more reliable signal.

Versus benchmark. Over the same window, SPY buy-and-hold returned ≈ **+38 %** over 2022-04 → 2025-12 † (external estimate, yfinance, not yet implemented in repo backtest). Strategy underperforms on raw return but trades only on selected events (≈ 3.5 % of bars). Risk-adjusted (Sharpe, exposure-weighted) is the meaningful comparison at this sample size.

When F1 and profitability disagree.

Maximising macro-F1 picks moderate barriers that classify well on small moves. Maximising return picks **`pt 2.0 / sl 3.0 / mh 20`**, fewer correct labels, but each one moves further. **The two optima do not coincide.**

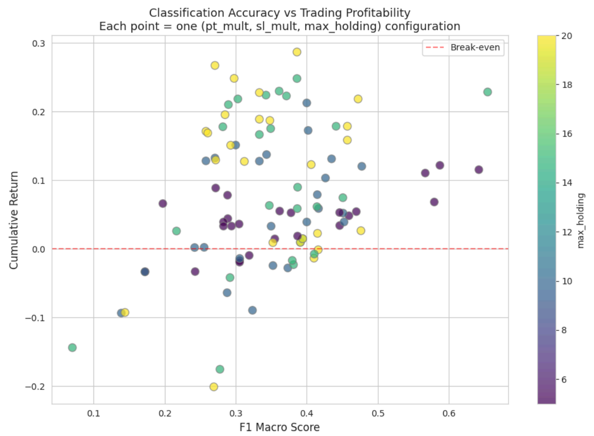

FIG 11.1. Each dot is one of the 100 (pt, sl, mh) configurations. The Pareto frontier lies along the upper-right edge of the cloud, no single point maximises both objectives.

F-β as a single dial. Conceptually, β < 1 weights precision (don't trade unless sure); β > 1 weights recall (don't miss the move). The thesis exposes the dial as a tunable; the cost-aware sweep across β is listed in the roadmap as the next step.

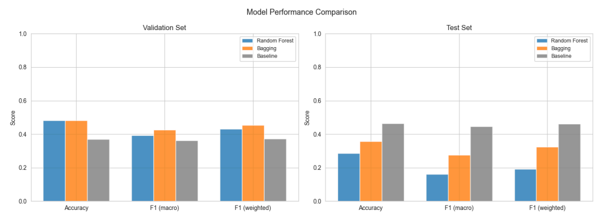

FIG 11.2. Four final configurations head-to-head. *Baseline* wins test accuracy by class-prior luck ($F1_m$ only .16); *Best for Profit* trails on F1 but holds `+18.7 % realised, +25.85 % oracle ceiling`. The augmented event set (detector + touch) reaches a **79 % win rate** at the profit-optimal config; the lift partly reflects that touch events fire more frequently than full-pattern completions.

KEY TAKEAWAYS

- ✓ **Event-based learning is viable.** A 3.5 %-of-bars subset gives a leakage-free training set.
- ✓ **Volume statistics dominate.** The model relies on feature interactions, not a single signal.
- ✓ **F1 ↔ profit gap demands two optima.** Reporting either alone misleads.
- ✓ **Bagging > RF on validation**; baseline wins on accuracy only; Bagging leads F1 (.275 vs .16).

CLOSING

What's solved, what's next.

Event-based ML, labelled by a triple barrier, is the right framing for this problem. The open question is no longer *does it work?* but *how cheaply does it scale?*

CONCLUSIONS

01 Event-based labelling works.

Geometric completion + triple-barrier outcomes produce a clean supervised problem.

02 Bagging is the right ensemble here.

Beats Random Forest on validation F1 (.425 vs .393) and on cumulative return.

03 F1 and profit are different objectives.

Reporting both is necessary; reporting only one is misleading.

04 Sample size dominates.

94 trainable events too few for stable test-set superiority; CV optima are the stronger signal.

WHAT WORKED VS WHAT WAS WEAK

WORKED

- Event-only training set
- Triple-barrier labelling
- Bagging best validation F1
- Walk-forward CV

WEAK

- Test-fold variance › signal
- F1 ⇔ profit disagreement
- volume_std overweighted

LIMITATIONS

- **Single asset.** SPY only, generalisation unverified.
- **Small sample.** 94 trainable events (142 detected); 28-event test has wide CIs.
- **No transaction costs modelled.** Slippage, commissions and borrow on shorts not yet implemented (thesis §19).
- **Touch-event noise.** Weaker structures lift recall, depress precision.
- **No regime gate.** HMM / vol-state overlay deferred.

ROADMAP

1. Multi-asset extension · FX, futures, crypto · pool events
2. HMM regime layer · vol/trend state conditioning
3. Cost-aware F-β tuning · pick β from spread/slippage
4. Detector probabilities · replace 0/1 with calibrated scores
5. Online retraining · rolling refit at every event close

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One line each.

Abbreviations and units used across the deck. For Q&A and offline reading.

MARKET & INDICATORS

SPY	SPDR S&P 500 ETF, U.S. large-cap stocks.
ATR	Average True Range. Volatility unit, 14-bar window.
σ_{ATR}	Barrier width unit. `pt · ATR` = TP distance.
RSI-14	Relative Strength Index, 14-bar momentum.
MACD	Moving Average Convergence / Divergence (EMA diff).
BB width / %B	Bollinger Band channel width / close position.
OBV	On-Balance Volume. Running signed-volume sum.
SMA	Simple Moving Average. Used in `sma_N_dist`.
OHLCV	Open, High, Low, Close, Volume.

ML, STATISTICS & PERFORMANCE

$F1_m$	Macro F1, harmonic mean of precision and recall.
F-β	F1 with adjustable P/R weight. $\beta \neq 1$.
Gini	Tree split criterion; Δ Gini = feature importance.
ANOVA F	Tests whether a feature mean differs across labels.
CV	Cross-validation; event-CV folds over events.
RF / Bagging	Tree ensembles. RF samples features; Bagging rows.
Triple barrier	First of $\{+pt\cdot\sigma, -sl\cdot\sigma, t + mh\}$ hit.
Sharpe	Per-trade Sharpe: mean(returns) / std(returns). *Not* annualised.
Profit factor	Σ winning trades / IΣ losing tradesI.
Max DD	Maximum drawdown from running peak.
HMM	Hidden Markov Model; deferred regime layer.

READING THE FIGURES

The **yellow diamond** marks the signal bar. Cyan / red triangles mark touch points used to fit the geometry.

REPO + THESIS

Code & verified numbers: `github.com/zaetae/regime-aware-ml-trading`. Source-of-truth: `PROJECT_THESIS_PRESENTATION.md`.

NOTEBOOK MAP

- `03` · Pattern validation & visual tuning
- `04` · Pattern structure validation (triangles)
- `05` · Triple-barrier labelling
- `06` · Channel detection & visualisation
- `07` · Data source comparison (yfinance ↔ Alpha Vantage)
- `08` · Comprehensive detector validation
- `09` · Feature engineering (38 indicators + binaries)
- `10` · Model training (RF, Bagging, baseline)
- `11` · Experiment summary & consolidation
- `12` · Hyperparameter optimisation & profitability
- `13` · Generalisation, F-β, profitability robustness
- `src/` · data, patterns, labelling, features, models, backtest